1. Differentiate Dynamic Test with Static Test.

Feature	Static Testing	Dynamic Testing
Definition	Testing **without executing** the code	Testing **with code execution**
Purpose	To find errors in **code, design, or documentation**	To find **runtime errors** and verify system behavior
Stage	Done **early** in the development cycle	Done **later**, after code is developed and runnable
Techniques Used	Code reviews, walkthroughs, inspections, static analysis	Unit testing, integration testing, system testing
Performed By	Developers, reviewers, QA teams	Testers or QA teams
Tools Example	SonarQube, ESLint, Checkstyle	Selenium, JUnit, TestNG
Detects	Syntax errors, coding standard violations, dead code	Logical errors, performance issues, UI functionality
Cost Effectiveness	More cost-effective (detects defects early)	Less cost-effective (bugs found later are costlier)

2. Briefly explain **Black Box Testing**.

   Black Box Testing is a **software testing method** where testers check the **functionality** of an application **without knowing its internal code or structure**.

3. An e-commerce system is built to handle the trading of computer accessories. The system processes 200000 transactions each week. The system is required to produce an analysis report according to the standard provided. Suggest the types of tests that are important for the system.

- Load Testing:
  - Checks system performance under normal and peak loads.
  - Ensures transactions and report generation work smoothly.
  - Identifies performance bottlenecks like slow response times or high CPU usage.
- Stress Testing:
  - Pushes the system beyond expected limits to find its breaking point.
  - Tests system stability during traffic spikes or server failures.
  - Ensures graceful recovery from crashes and extreme loads.

4. Relate the types of testing to the following keywords:
   a. Down time
   b. Response Time
   c. User interface
   d. Workload

Keyword	**Related Testing Type**	**Explanation**
Down time	Reliability Testing / Recovery Testing	Measures how long the system stays operational before failure and how it recovers after failure. It ensures system availability.
Response Time	Performance Testing	Evaluates how fast the system responds to user or system inputs, especially under load.
User Interface	Usability Testing	Checks the layout, visual elements, ease of use, and correctness of the user interface to ensure a good user experience.
Workload	Load Testing / Stress Testing	Simulates multiple users or high volume of data to see if the system can handle the expected (or beyond) usage levels without breaking down.

4. Compare **Volume Test, Stress Test, Load Test,** and **Performance Test**

- **Volume Test**
  - is tested with a large number of data to evaluate its performance, stability, and scalability
- **Stress Test**
  - beyond normal operational capacity, often to a breaking point
- **Load Test**
  - to understand the behavior of the application under a specific expected load
- **Performance Test**
  - to determine how fast some aspect of a system performs under a particular workload

5. In what situation would an Exploratory Test can be applied? Explain your answer.

Exploratory Testing is ideal in situations where:

- **The application is critical and time is limited.** When a software application must be delivered quickly, such as in a tight release schedule or a high-stakes environment. Exploratory testing allows testers to rapidly identify issues without waiting for fully written test cases. It provides immediate feedback on system behavior.

- **Early iterations are required.** In the early stages of development, when requirements may still be evolving and detailed documentation is limited, exploratory testing enables testers to learn about the system as they test. This approach helps uncover unexpected behaviors and edge cases that scripted tests might miss.

6. Differentiate **Alpha Testing** and **Beta Testing**.

Criteria	**Alpha Testing**	**Beta Testing**
Site	Developer's site	Customer's site
Testers	Internal employees, developers, and testers.	Real users or selected external testers.
Environment	Conducted in a **lab/test environment**.	Conducted in a **real-world environment**.
Scope	Focuses on **functionality, usability, and performance**.	Focuses on **user experience, real-world performance, and final feedback**.

7. What is Error Guessing?

Error Guessing is a software testing technique where testers use their experience, intuition, and past knowledge to predict areas where bugs are likely to occur. Instead of following a structured test case, testers guess potential errors based on common mistakes and known weaknesses.

8. A web application is developed to determine the loyalty types of a club member. If the member stays in the club less than 12 months, the member will be awarded as Normal grade, VIP grade will be awarded to whom has become member between 1 to 3 years. VVIP grade is for those members who stay more than 3 years.

   Decide and apply appropriate Black Box Testing techniques to determine the number of test cases. You must show the detailed working.

Answer:

Parameter	Equivalence Class	Representative Value	Result
Duration as a member (x)	vEC1: $0<x<1$	0.5	Normal
	vEC2: $1<= x <= 3$	2	VIP
	vEC3: $x > 3$	6	VVIP
	ivEC1: $x <= 0$	-5	Error
	ivEC2 : x is unrealistic number	1000	Error

Test Cases:
TC1: Test with valid value for Normal member. Input value (test value) is 0.5.
TC2: Test with valid value for VIP member. Input value (test value) is 2.
TC3: Test with valid value for VVIP member. Input value (test value) is 6.
TC4: Test with invalid value where duration is less than or equal to 0. Input value (test value) is -5.
TC5: Test with invalid / unrealistic value. Input value (test value) is -1000.

9. A special machine is used to check and categorise the sumo wrestler contestant, there are three classes that a man can choose to join. The machine will check on the weight of the contestant. For the Heavyweight class, the man must have at least 254 lbs and above. For Middleweight, the weight of the wrestler must be between 188 lbs and 253 lbs. If a man's weight is 187 lbs and below, he can join the Lightweight class.

- Using Equivalence Class Partitioning, create a table to determine the equivalence class, representative value, parameter and expected result.
- Identify and list the test case using Boundary Value Analysis.

Answer:

Parameter	Equivalence Class	Representative Value	Result
Contestant weigh, (x)	**vEC1: 137 <= x <= 187**	140	Light Weight
	vEC2: 188<= x <= 253	6	Middle Weight
	vEC3 : x >= 254	255	Heavy Weight
	ivEC1: x < 137	5	Error
	ivEC2 : x is unrealistic number	100000	Error

** Take note that for vEC1, there should be a minimum weight for a human, the example given is the minimum weight.

10. In order for a student to graduate from a Bachelor Degree, the student must clear all payments before the final examination. The CGPA of the student to pass is at least 2.67 point and the student must fill in the feedback form before graduation. If the student does not fill in the feedback form, the student will be prohibited from graduating and the student will be requested to pay 50 dollars for additional charges.
    - Construct the Cause Effect Graph.
    - From the Cause Effect Graph, create a Decision Table.

Answer:

Condition	TC1	TC2	TC3	TC4	TC5
Clear Payment	T	F	-	-	
CGPA > 2.67	T	-	F	-	
Filled Feedback Form	T	-	-	F	T
Action (Effect)					
Allow graduate	Y				
Cannot graduate		Y	Y	Y	
Charge 50					Y

11. AGV cinema in Malaysia has given free tickets or 50% discount to the public for the celebration of Independence Day. The cinema is using a ticket printing system to print the ticket. In order for the public who is eligible for the promotion, the person must show his Identity Card at the ticket counter. For those people who come along with his son/daughter, maximum 4 free tickets are given to each family. If the person does not come along with his son/daughter but with his wife/partner, a 50% discount is given to each ticket. For those elderly people whose age is 60 and above, free tickets will be given to them with the presence of their identity card.

- Construct the Cause Effect Graph.
- From the Cause Effect Graph, create a Decision Table.

Answer:

Decision Table	**TC1**	**TC2**	**TC3**	**TC4**	**TC5**
Conditions					
Age ≥ 60	Y	N	N	N	N
With Son/Daughter	N	Y	N	N	N
With Wife/Partner	N	N	Y	N	N
Show Identity Card	Y	Y	Y	N	Y
Actions					
Free Ticket	Y	Y	N	N	N
Max 4 Free Tickets	N	Y	N	N	N
50% Discount	N	N	Y	N	N
No Promotion	N	N	N	Y	N

TC1: Elderly person (≥60) with an identity card → Free Ticket

TC2: Person with a son/daughter and an identity card → Max 4 Free Tickets

TC3: Person with a wife/partner and an identity card → 50% Discount

TC4: No valid conditions met (identity card not shown) → No Promotion

TC5: Edge case: identity card shown but none of the main conditions met → No Promotion